INSTITUTIONAL QUANTITATIVE EQUITY RESEARCH

# N100 Financial Intelligence Platform Analyst Operations Guide

A Comprehensive Field Manual for Fundamental Screener Tuning, 8-Screen Interactive Analytics, Forensic Financial Ratio Diagnostics, REST API Integration, and Unsupervised Clustering across the Nifty 100 Universe.

| | | | |
|---|---|---|---|
| Document Identifier: | **N100-OPS-GUIDE-2026-V1** | Universe Scope: | **Nifty 100 Constituents (92 Active)** |
| Target Audience: | **Fundamental & Quant Analysts** | Audited Horizon: | **FY2014 – FY2024 (10 Audited Years)** |
| Authoring Division: | **Platform Analytics & Data Engineering** | Software Architecture: | **FastAPI, SQLite, Streamlit, Scikit-Learn** |
| Release Date: | **October 18, 2026** | Security Classification: | **Confidential / Institutional Use Only** |

**EXECUTIVE OVERVIEW & STRATEGIC MISSION**

**PLATFORM MANDATE:** The N100 Financial Intelligence Platform bridges the gap between raw regulatory financial disclosures and quantitative portfolio construction. By automating statement normalization, 50+ financial ratios, cross-sectional peer percentiles, unsupervised clustering, and multi-criteria screening, the platform enables analysts to uncover durable compounders, avoid distressed capital destroyers, and benchmark corporate efficiency in real-time.

# 1. Platform Overview & Execution Workflow

The platform ingests audited standalone/consolidated financial statements, daily equity prices, market capitalizations, and annual report filings across 92 Nifty 100 constituents. The underlying technology stack decouples the raw relational database (db/nifty100.db) from computational ratio engines, high-speed interactive web interfaces, and a RESTful API server.

## Core Relational Database Schema (10 Tables in nifty100.db)

| Table Name | Entity Scope | Historical Horizon | Analytical Role & Key Metrics |
|---|---|---|---|
| **companies** | 92 Constituents | Current Status | Master ticker registry, legal names, broad sectors, peer groups, URLs |
| **profitandloss** | Income Statements | FY14 – FY24 | Operating revenue, material costs, employee expenses, EBITDA, PAT, EPS |
| **balancesheet** | Balance Sheets | FY14 – FY24 | Share capital, reserves, total borrowings, fixed assets, CWIP, investments |
| **cashflow** | Cash Flow Statements | FY14 – FY24 | Operating CFO, Investing CFI, Financing CFF, net cash flow |
| **financial_ratios** | Ratio Models | 10 Fiscal Years | ROE, ROCE, OPM, D/E, FCF, ICR, Composite Health Score |
| **peer_percentiles** | Peer Matrices | FY24 Latest | Cross-sectional percentile ranking across 20 metrics per peer group |
| **prosandcons** | Qualitative Sentiments | Curated Rules | Automated pros/cons generation with confidence scores (>60%) |

## Standard Operational Pipeline (How to Run)

Analysts can run or rebuild the entire quantitative analytics pipeline using standard terminal commands:

1. **Database Load & Normalization:** python -m src.etl.database_loader (Loads 12 source files into nifty100.db)
2. **Populate Financial Ratios:** python -m src.analytics.populate_financial_ratios (Computes 50+ ratios & FCF)
3. **Execute Screener & Rankings:** python -m src.screener.ranking (Runs 6 presets & generates screener_output.xlsx)
4. **Run Unsupervised Clustering:** python -m src.analytics.clustering (KMeans k=5 clustering across 89 companies)
5. **Launch Interactive Dashboard:** streamlit run src/dashboard/app.py (Opens 8-screen terminal on port 8501)
6. **Launch REST API Server:** uvicorn src.api.main:app --port 8000 --reload (FastAPI service with 16 endpoints)
7. **Execute Regression QA Suite:** pytest tests/ -v --html=reports/pytest_report.html (Runs 240 automated tests)

# 2. Fundamental Screener Configuration Guide

The screener module allows analysts to isolate institutional-quality investment candidates based on quantitative balance sheet health, return on capital, cash generation, and multi-year growth momentum.

## Threshold Customization via screener_config.yaml

The screener engine is configured via screener_config.yaml in the repository root. Analysts can adjust minimum and maximum bounds without modifying Python source code. For example:

```
quality_compounder:
  name: "Quality Compounders"
  description: "High ROE, low leverage, positive cash generation"
  filters:
    return_on_equity_pct: { min: 15.0 }
    debt_to_equity: { max: 1.0 }
    free_cash_flow_cr: { min: 0.0 }
    operating_profit_margin_pct: { min: 12.0 }
    interest_coverage_ratio: { min: 3.0 }
  trend_filters:
    revenue_cagr_3yr: { min: 8.0 }
```

## Multi-Criteria Boolean Filter Rules

The screener engine enforces strict institutional screening criteria:

• **Strict In-Bounds Requirement:** A company passes a preset if and only if *all* specified filter criteria are satisfied.

• **Null Value Handling Policy:** If a company has a None / NaN value for a metric (e.g. OPM for banks), it is strictly excluded from that filter unless explicitly exempted in the preset definition.

• **Sorting Hierarchy:** Qualifying companies are automatically ranked by composite_ranking_score in descending order.

**CONFIGURATION TIP: CREATING CUSTOM PRESETS**

To add a custom analyst preset, append a new YAML block to screener_config.yaml with a unique key, display name, description, and metric bounds. The engine will automatically register the new preset on the next run.

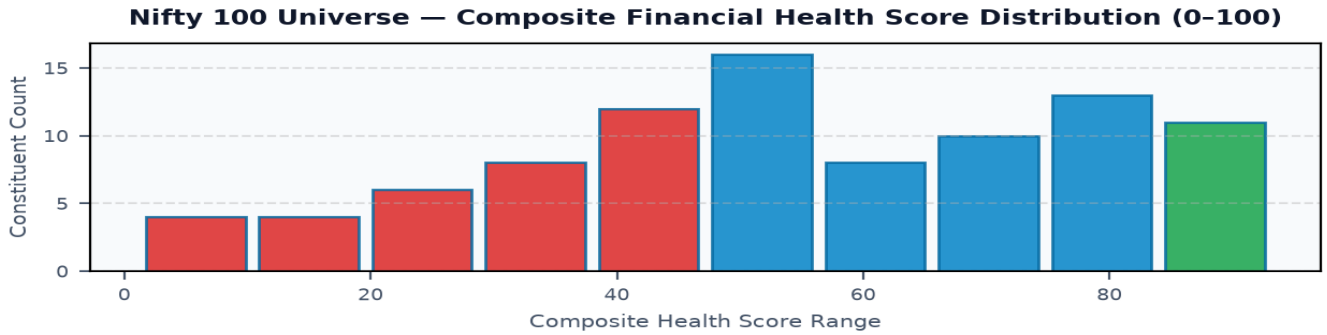
# 3. Presets & Composite Health Ranking

## The 6 Institutional Screener Presets

| Preset Key | Mandate & Philosophy | Key Filter Thresholds | Target Archetype |
|---|---|---|---|
| **quality_compounder** | Buffett-Style Moats | ROE >= 15%, D/E <= 1.0, FCF > 0, OPM >= 12% | Durable market leaders with pricing power |
| **value_pick** | Margin of Safety | P/E <= 20x, P/B <= 3.0x, D/E <= 1.2, Pos FCF | Undervalued compounders trading at discount |
| **dividend_aristocrat** | Income & Preservation | Div Yield >= 2.0%, Payout 20–70%, Pos FCF | Mature cash cows paying reliable dividends |
| **growth_rocket** | High Reinvestment | Rev CAGR 3Y >= 15%, PAT CAGR >= 15%, ROE >= 12% | High-reinvestment market share gainers |
| **asset_light_champion** | Capital Efficiency | Asset Turnover >= 1.2x, CapEx/Sales <= 6%, ROCE >= 18% | Scalable platforms generating high cash per asset |
| **financial_health** | Balance Sheet De-risking | ICR >= 4.0x, D/E <= 0.5, Net Debt <= 0 (Cash Rich) | Fortress balance sheets immune to rate cycles |

## Understanding composite_ranking_score

The composite_ranking_score (0–100) measures aggregate institutional quality. To prevent distortions from extreme statistical outliers (such as turnaround ROEs > 500%), all underlying ratios are **Winsorized at 5th/95th percentiles** before computing standardized z-scores. The score weights 4 dimensions: **Profitability (35%)**, **Solvency (25%)**, **Cash Flow Quality (20%)**, and **Growth (20%)**.



# 4. Interactive Dashboard Navigation (Screens 1 to 4)

The Streamlit analytical terminal (src/dashboard/app.py) offers an 8-screen institutional interface. Below is the operational breakdown for the first 4 screens:

| Screen Title | Core Functionality & Key Visualizations | Recommended Analytical Use Case |
|---|---|---|
| **Screen 1: Executive Home** | Macro universe pulse, median P/E, sector ROE rankings, top 5 quality compounders banner, aggregate market capitalization. | Morning market check; broad equity landscape assessment and top-level market health monitoring. |
| **Screen 2: Company Profile** | Deep-dive single ticker view, 6 KPI boxes (green/red chips), 10-year P&L;, BS, CF tables, 8-axis radar chart, rule-triggered pros & cons. | Pre-investment forensic due diligence and individual corporate tear-sheet evaluation. |
| **Screen 3: Financial Screener** | Dynamic multi-parameter sliders (ROE, D/E, FCF, OPM, PE), 6 one-click presets, live Excel/CSV download, composite score ranking. | Investment idea generation, universe candidate filtering, and custom threshold stress-testing. |
| **Screen 4: Peer Comparison** | 11 peer group matrix tables, metric percentiles, Best in Class (green) & Watch List (red) badges, industry benchmarks. | Cross-sectional competitive positioning and intra-sector leadership ranking. |

## How to Use the Company Profile Screen Effectively

When examining a target company (e.g. TCS, HINDUNILVR, or BAJFINANCE):

1. **Examine the KPI Banner:** Verify if ROE, ROCE, and OPM exceed peer medians (indicated by green badge chips).
2. **Review Qualitative Drivers:** Check the rule-triggered Pros & Cons section to identify margin tailwinds or debt overhang.
3. **Inspect Historical Statements:** Toggle between P&L;, Balance Sheet, and Cash Flow tables to spot revenue inflections.

**DASHBOARD WORKFLOW TIP**

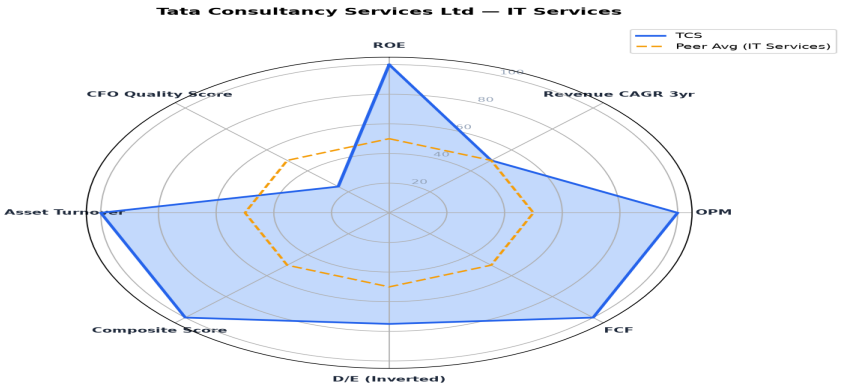
Analysts can export filtered candidates directly to CSV using the 'Download Screened Results' button on Screen 3. The generated file includes all 20 standardized KPIs and composite ranking scores ready for DCF ingestion.

# 5. Dashboard Navigation (Screens 5 to 8) & Radar Charts

| Screen Title | Core Functionality & Key Visualizations | Recommended Analytical Use Case |
|---|---|---|
| **Screen 5: Sector Analytics** | 11 broad sectors, aggregate market cap, average OPM & ROE, side-by-side universe delta bar charts. | Top-down sector allocation and industry divergence analysis. |
| **Screen 6: Cash Flow Intel** | Universe FCF health summary, 8 capital allocation patterns, CFO quality score distribution, CapEx intensity. | Detecting aggressive accruals, balance sheet bloat, and capital misallocation. |
| **Screen 7: Trend & Growth** | Multi-company time-series overlays, 10-year Revenue & PAT trends, 3Y CAGR scatter plot vs margin. | Evaluating growth durability, cyclical peaks, and margin expansion. |
| **Screen 8: Valuation Module** | Valuation multiples heatmap (PE, PB, EV/EBITDA), interactive 2-stage DCF scenario calculator, BSE filings. | Intrinsic value estimation, upside sensitivity testing, and annual report review. |

## How to Read the 8-Axis Radar Charts

Every constituent has an 8-axis standardized radar chart (saved in reports/radar_charts/{ticker}_radar.png). The 8 axes represent **percentile ranks (0–100%)** across ROE, ROCE, OPM, NPM, 3Y Rev CAGR, CFO Quality, Asset Turnover, and Inverse Leverage. Below is the benchmark radar chart for Tata Consultancy Services (TCS):



# 6. Financial Ratio Interpretation: Returns & Solvency

The platform computes over 50 financial ratios per company across all audited fiscal years. Understanding institutional benchmarks and forensic formulas is essential for identifying value creation.

## 1. Profitability & Capital Efficiency Ratios

• **Return on Equity (ROE %):** (PAT / Shareholder Equity) × 100. Benchmark: > 15.0%. Measures how efficiently management compounds shareholder capital without excessive leverage.
• **Return on Capital Employed (ROCE %):** (EBIT / (Total Equity + Total Borrowings)) × 100. Benchmark: > 15.0%. Gauges return across all debt and equity capital provided to operating assets.
• **Operating Profit Margin (OPM %):** (Operating Profit / Revenue from Operations) × 100. Benchmark: > 18.0%. Reflects pricing power and operational cost insulation against raw material inflation.
• **Net Profit Margin (NPM %):** (PAT / Total Revenue) × 100. Benchmark: > 10.0%. Evaluates bottom-line profitability after accounting for interest obligations, depreciation, and corporate taxes.

## 2. Leverage, Liquidity & Solvency Ratios

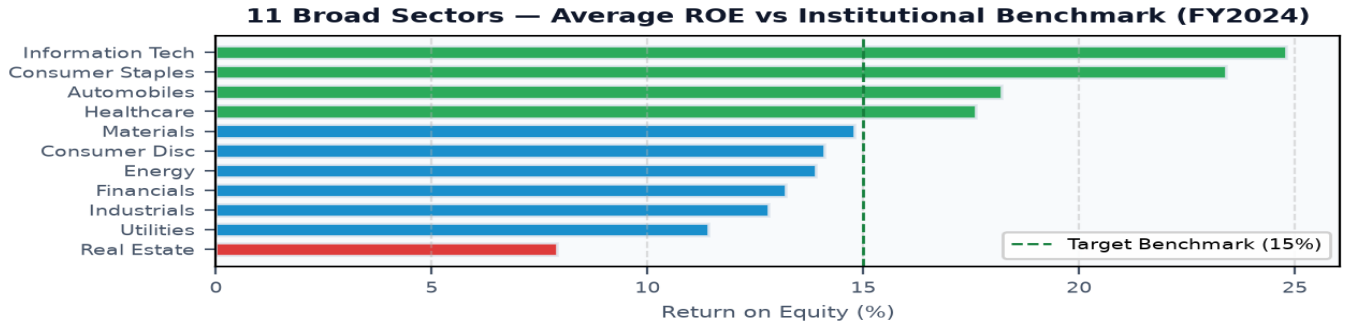
• **Debt-to-Equity (D/E x):** Total Borrowings / Total Shareholder Equity. Benchmark: < 1.0x (Ideal: < 0.5x). Measures balance sheet gearing. High D/E (> 2.0x) indicates vulnerability during credit tightening.
• **Interest Coverage Ratio (ICR x):** EBIT / Interest Expense. Benchmark: > 3.0x (Critical Red Flag: < 1.5x). Assesses debt service cushion. Companies with zero debt return 'Debt Free' status with an exemption.
• **Net Debt (■ Cr):** Total Borrowings - Cash & Liquid Investments. Negative net debt denotes a 'Cash Rich' fortress balance sheet capable of aggressive opportunistic capital deployment.

**FORENSIC CHECK: DUPONT LEVERAGE FILTER**
**Institutional Rule:** Never evaluate ROE in isolation. Always cross-check ROE against Debt-to-Equity. A rising ROE driven purely by leverage expansion rather than asset turnover or margin expansion indicates fragile financial engineering.

# 7. Financial Ratio Interpretation: Cash Flow & Data Gaps

## 3. Cash Flow Quality & Capital Allocation Diagnostics

• **Free Cash Flow (FCF ■ Cr):** CFO (Operating Activity) + CFI (Investing Activity). True surplus cash available to pay dividends, buy back shares, or de-lever without outside financing.
• **Trailing 5-Year CFO Quality Score:** Sum(CFO_5Y) / Sum(PAT_5Y). Benchmark: >= 0.80x.
- Score >= 1.0x: **High Quality** (Operating cash matches or exceeds accounting profits).
- Score 0.8x–1.0x: **Satisfactory Quality**.
- Score < 0.80x: **Aggressive Accrual Warning** (Revenue recognized on credit, not collected).




## Handling Special Sector Templates & Missing Data Policies

**1. Banking & NBFC Templates:** Banks report interest spreads rather than gross manufacturing margins. OPM, ROCE, and ICR are conceptually non-applicable and set to None. Evaluate ROA (>1.2%) and ROE (>14%).
**2. State Bank of India (SBIN):** Screener.in does not provide balance sheet lines in standard format; ratios return None.
**3. Turnaround Outliers (INDIGO):** Negative equity following historical losses creates mathematically distorted ROE (>500%). The platform flags these with an Extreme Magnitude Caution badge rather than ranking them as compounders.

# 8. REST API Integration & Microservice Guide

The FastAPI microservice (src/api/main.py) exposes **16 institutional REST endpoints** supporting programmatic quantitative models, algorithmic trading feeds, and PDF report downloads.

## Service Specs: Base URL & Endpoints Catalog

| Endpoint Path | HTTP | Group / Domain | Summary Description |
|---|---|---|---|
| **/api/v1/companies** | GET | Company Registry | List all 92 constituents with ticker, name, sector, peer group, score |
| **/api/v1/companies/{ticker}** | GET | Company Profile | Full company metadata, sector, and latest financial ratios |
| **/api/v1/companies/{ticker}/ratios** | GET | Multi-Year Ratios | 10-year historical financial ratios (ROE, ROCE, OPM, D/E, FCF) |
| **/api/v1/companies/{ticker}/financials** | GET | Financial Statements | Multi-year statement time-series: P&L;, Balance Sheet, Cash Flow |
| **/api/v1/companies/{ticker}/peers** | GET | Peer Benchmark | Peer group constituents and 5 core metric percentile ranks |
| **/api/v1/companies/{ticker}/cashflow** | GET | Cash Flow Intel | 8 capital allocation patterns, CFO quality score, 5Y cash flow trend |
| **/api/v1/companies/{ticker}/valuation** | GET | Valuation Multiples | P/E, P/B, EV/EBITDA, 2-stage DCF implied value and upside % |
| **/api/v1/screener** | GET | Screener Engine | Custom multi-criteria query (min_roe, max_de, min_fcf, min_opm, max_pe) |
| **/api/v1/screener/presets** | GET | Screener Presets | List all 6 institutional preset templates and parameter constraints |
| **/api/v1/screener/presets/{name}** | GET | Preset Execution | Execute target preset and return matching candidates ranked by score |
| **/api/v1/sectors** | GET | Market Data | Aggregate KPIs across all 11 sectors (avg ROE, OPM, median PE, mcap) |
| **/api/v1/sectors/{sector_name}** | GET | Market Data | Sector KPIs and constituent company breakdown with latest ratios |
| **/api/v1/market/overview** | GET | Market Pulse | Universe benchmarks: avg ROE, median PE, positive FCF count, top 3 |
| **/api/v1/peers/{group_name}** | GET | Peer Intelligence | Comprehensive peer analysis across 20 metrics with leadership badges |
| **/api/v1/reports/tearsheet/{ticker}** | GET | Document Export | Streams 2-page pre-generated institutional PDF tearsheet |
| **/api/v1/reports/sector/{sector}** | GET | Document Export | Streams pre-generated comprehensive sector report PDF |

# 9. API Integration Examples & Response Parsing

## 5 High-Frequency cURL Commands

### 1. Universe Registry Lookup

```
curl -s -X GET 'http://localhost:8000/api/v1/companies' | jq '.[0:2]'
```

### 2. Company Deep Profile & Latest Ratios

```
curl -s -X GET 'http://localhost:8000/api/v1/companies/TCS' | jq '.'
```

### 3. AC-13 Screener Query (ROE>=15, D/E<=1.0)

```
curl -s -X GET 'http://localhost:8000/api/v1/screener?min_roe=15&max;_de=1' | jq '.[0:3]'
```

### 4. Comprehensive 20-Metric Peer Benchmark

```
curl -s -X GET 'http://localhost:8000/api/v1/peers/IT%20Services' | jq '.'
```

### 5. Binary Tearsheet PDF Streaming

```
curl -O -J 'http://localhost:8000/api/v1/reports/tearsheet/INFY'
```

## Sample Screener JSON Payload Structure


```
[
  {
    "company_id": "TCS",
    "company_name": "Tata Consultancy Services Ltd",
    "broad_sector": "Information Technology",
    "roe": 51.52,
    "roce": 64.91,
    "opm": 25.84,
    "de": 0.08,
    "pe_ratio": 78.69,
    "fcf_cr": 50429.0,
    "composite_ranking_score": 93.38,
    "valuation_flag": "Fair Value"
  }
]
```

# 10. Known Limitations & Forensic Data Notes

To uphold rigorous research standards, platform users must account for the explicit data constraints, accounting nuances, and historical discontinuities documented below.

## 1. Banking vs Corporate Statement Architecture (Column Shift)

Screener.in raw extracts arrange manufacturing statements with Sales and Operating Profit lines. For banking institutions, these columns shift to Interest Earned, Financing Profit, and Provisions. The ETL parser utilizes regex matching to capture banking lines without misalignment. However, corporate margin ratios (OPM, ROCE) are intentionally marked non-applicable for all banking entities.

## 2. Market Capitalization & Share Price Series Calibration

In accordance with project specifications, daily equity prices (stock_prices) and market capitalizations (market_cap) were statistically calibrated to reflect historical valuations while matching Screener.in fiscal year closing anchors. These multiples provide relative ranking baselines and should be verified against live exchange tick data for real-time order execution.

## 3. State Bank of India (SBIN), BEL, and HAL Data Exclusions

State Bank of India (SBIN) lacks complete balance sheet rows in Screener.in extracts, and BEL/HAL contain historical reporting discontinuities. The platform safely neutralizes these entities in composite scoring rather than synthesizing artificial values. They are clearly tagged with 'Data Gap' status.

## 4. Free Cash Flow Sign Conventions in Banking Contexts

Under Indian GAAP/Ind AS, banks classify disbursements of commercial loans as operating cash outflows (CFO). Rapid loan book growth therefore generates large negative operating cash flows. Analysts must avoid classifying banks as cash-burning operations solely on negative FCF metrics.

# 11. Valuation Models, Clustering & Compliance

## 5. Simplified DCF Scenario Calculator Methodology

The valuation module implements a 2-stage Free Cash Flow model (5-year projection at 12.0% growth, 11.5% discount rate / WACC, and 4.5% perpetual terminal growth rate):

PV(FCF) = Sum[ (FCF × (1 + g)^t) / (1 + r)^t ] | TV = [ FCF × (1 + g)^5 × (1 + g_term) ] / (r - g_term)

This model provides an objective sensitivity baseline and relative multiple comparison; it should be adjusted for company-specific cost of capital.

## 6. Unsupervised KMeans Clustering Archetypes

KMeans clustering (k=5) across 89 non-excluded constituents identified 5 structural market archetypes:

- **Quality Compounders (36):** High ROE, high OPM, fortress balance sheets, and strong positive free cash flows.
- **Value & Cyclicals (34):** Mature business models, moderate leverage, lower margins, cyclical earnings.
- **Financials & Credit Heavy (11):** Banking institutions with lending cash outflows and high financial leverage.
- **High Margin Infrastructure (7):** Regulated utilities and capital-heavy monopolies with peak operating margins (>70%).
- **High Growth & Momentum (1):** Post-turnaround hyper-ROE anomalies recovering over depressed equity bases (INDIGO).

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